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ATTACHMENT I

NATIONAL WEATHER SERVICE

ENGINEERING and ACQUISITION BRANCH

SPECIFICATION FOR AEROLOGICAL SOUNDING BALLOONS

U.S. DEPARTMENT OF COMMERCE
NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION
NATIONAL WEATHER SERVICE
Silver Spring, Maryland 20910-3283

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AEROLOGICAL SOUNDING BALLOONS
Revision L
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AEROLOGICAL SOUNDING BALLOONS

1 INTRODUCTION

1.1 Background

The National Oceanic and Atmospheric Administration (NOAA)'s National Weather Service (NWS) uses 78,000 General Purpose 26 km (GP-26) and GP20 km aerological sounding balloons annually to support NWS radiosonde observations required by the Office of Observations (OBS) and its NWS Weather Forecast Offices. A aerological sounding balloon is a balloon used to carry radiosondes aloft. Radiosondes measure atmospheric temperature, humidity, pressure, wind speed, and wind direction from the surface to an altitude of approximately 30,000 meters. The radiosonde observations are a critical source of meteorological data for these weather forecast models. The NWS radiosonde observations provide atmospheric observations for development of watches, warnings, weather forecasts and aviation support. The NOAA/NWS publishes 3-hour, 6-hour, 12-hour, 24-hour, and 48-hour upper level atmospheric models for watches, warnings, forecasts, and aviation support.

The aerological sounding balloons are launched at all 102 NWS Upper-Air sites and thus must survive a wide variety of launch and flight weather conditions, while providing an average ascent rate of 5 m/s for accurate sensor measurements. The aerological sounding balloon performance is critical to accurate sensor readings on the supported radiosonde observations. The aerological sounding balloon performance specification provides the radiosonde sensors with optimal ascent rate and burst altitude necessary for reliable and accurate radiosonde observations.

1.2 Scope

This specification defines the NWS aerological sounding balloons performance requirements. Appendix A describes tests used to evaluate the balloons.

1.2.1 Balloon Designations.

Balloon types are designated as General Purpose (GP) followed by a two digit numeric suffix designating their minimum bursting height in kilometers (km) above sea level. For example, GP20 is the designation for the General Purpose type balloon with a minimum bursting altitude of 20.0 km. The balloon designations are in TABLE I.

TABLE I. Balloon Designations

DESIGNATION	NATIONAL STOCK NO.	AGENCY STOCK NO.
GP20	NWS1-20-960-0001	J352-3

1.2.2 Specification Paragraph Titles.

The paragraph titles used in this specification are for convenience. The paragraph titles shall not be construed to imply that all requirements related to the title are contained in the paragraph.

2 APPLICABLE DOCUMENTS

2.1 General.
The following documents form a part of this specification to the extent specified herein.

2.2 Order of Precedence.
In the event of conflict between this specification and documents referenced, this specification shall take precedence. In the event of conflict with the body of this specification and the appendices, the body shall take precedence. The applicable document revision of a reference document for this contract shall be the revised document in effect on the day of release of the solicitation for this contract. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

2.3 Government documents.

2.3.1 Toxicology.

United States Center for Disease Control Agency for Toxic Substance and Disease Registry (ATSDR) Minimal Risk Levels (MRL) for Hazardous Substances

Available from: Agency for Toxic Substance and Disease Registry
ATSDR/OIRM,
1600 Clifton Road (E29),
Atlanta, Georgia 30333
USA http://www.atsdr.cdc.gov/mrls/index.html

2.3.2 Federal Aviation Administration.

Federal Aviation Requirement Part 101 (14 CFR 101.31 to 101.39)

Available from: Federal Aviation Administration
800 Independence Ave, SW. Room 700W, AJW-281
Attn: NAS Documentation Control Center (DCC)
Washington, DC.
20591 http://www.gpoaccess.gov/cfr/index.html

2.3.3 Federal Meteorological Handbook.

Federal Meteorological Handbook No. 1 (FCM-H1-2005)

Available from: Office of the Federal Coordinator for Meteorology
8455 Colesville Road, Suite 1500
Silver Spring, MD
20910 http://www.ofcm.gov/fmh-1/fmh1.htm

2.3.4 National Weather Service Manual 10-1401.

Operations and Services Upper Air Program NWSPD 10-14 Rawinsonde Observations, dated

June 2010
Available from: Office of Climate, Water, and Weather Services
1325 East West Hwy, Room 14201
Silver Spring, MD
20910 http://www.nws.noaa.gov/directives/sym/pd01014001curr.pdf

2.4 Paragraph Referencing.

A citation to a paragraph of this specification or a referenced document shall include all subordinate paragraphs of the cited paragraph.

2.5 Statistical Requirements.

Statistical requirements apply to each balloon type manufactured by each balloon manufacturer. A manufacturer cannot average the results of multiple balloon types to meet the specification statistical requirements.

2.6 Definitions.

Valid Flight	For the purpose of balloon requirements and testing, a valid flight is one where either a) the balloon meets its performance requirements regardless of the cause of the flight termination or b) a properly launched flight is terminated by balloon burst and the flight environment conditions are within the requirements of this specification regardless of whether the balloon has met its performance requirements.
Invalid Flight	For the purposes of balloon requirements and testing, an invalid flight is one where either a) the radiosonde or ground station fails prior to the balloon reaching its minimum burst height or b) the balloon struck some object in the launch process or c) the flight environment conditions are outside the requirements of this specification.
Flight Test Performance Percentage	The flight test performance percentage is 100 times the ratio of the number of balloons meeting their performance requirements to the total number of valid flights.
Free Lift	Free lift is the number of grams of lift available over and above the number of grams of lift required by a balloon to support the weight of a complete radiosonde train.
Nozzle Lift	Nozzle lift is the number of grams of free lift plus the grams of lift required by a balloon to support the weight of a complete radiosonde train excluding the weight of the balloon. The radiosonde train not including the weight of the balloon consists of the radiosonde, parachute, light stick, dereeler or train regulator, and twine.

Gross Lift	Gross lift is the nozzle lift plus the grams of lift required to support the weight of the balloon.
Nominal	Test samples performance is in accordance with these requirements.
Deficiency	A deficiency is an inadequate performance or quality of sample(s) as a result of a defect, design or manufacturing. Examples of deficiencies include, but are not limited to: samples exhibiting out-of-box defects, poor/inconsistent manufacturing quality,
Suspended	Deficiencies have been encountered that prevent continuation of the test, but are not severe enough to cause termination. The Contracting Officers Representative (COR) or Test Review Committee (TRC) will determine if a test should be suspended. A potential contractor shall have no more than four (4) weeks to correct the deficiencies and resume the test. Otherwise, a termination test status may result. Examples of suspension conditions include, but are not limited to: Deficient performance in a limited portion of the release / flight conditions identified in Tables 3.2.2 and 3.2.3. More than the minimally acceptable percentage of physical defects identified in section 3.5.
Termination	Results from test samples that have failed one or more tests with such a severity of deficiencies that to continue testing would not be in the Government's interest. Only the COR will make this determination and report the findings to the CO. Some examples for test Termination include, but are not limited to; samples failing a retest of a previous deficiency or samples with performance percentages below those identified in Table 3.3.7. Deficiencies that the manufacture is unable or unwilling to address.

3 REQUIREMENTS

3.1 General.

a) Aerological sounding balloons supplied in accordance with this specification shall meet all requirements herein. Balloons shall be capable of being safely and successfully released by one person from outside the inflation building and from a launch shelter in the environmental conditions of 3.2.2.

b) All requirements of this specification shall apply to all aerological sounding balloon types and designations except for those requirements specifically identified as applying to a particular aerological sounding balloon type or designation.

3.2 Environmental Conditions.

The aerological sounding balloons shall meet all specification requirements when subjected to all combinations of the encountered environmental conditions.

3.2.1 Transportation and Storage Environment Conditions.

Aerological sounding balloons shall conform to the requirements of this specification after exposure to the environmental conditions for transportation and storage.

3.2.1.1 Transportation Environment.

Aerological sounding balloons, when transported for up to 2 weeks in the Contractor-provided carton packaging during the 16 months after Government acceptance, shall conform to meet all requirements of this specification after exposure to all combinations of the following conditions:

TABLE II. Transportation Environment

Condition	Range
Air Temperature	-60.0 °C to +60.0 °C
Air Relative Humidity	5.0 % to 100.0 % including condensation
Dust	1 million particles per cubic meter, various sizes
Altitude	-50.0 m to 12.0 km, MSL
Radiation	Solar radiation at an altitude of 3.0 km
Vibration and Mechanical Shock	Conditions encountered during transportation by commercial carrier.

The balloons and their packing shall arrive at the National Logistics Support Center (NLSC) without shipping damage to enable NLSC to reship the balloons by only remarking the exterior carton.

3.2.1.2 Storage Environment.

Aerological sounding balloons, when stored in the Contractor-provided individual packaging, shall conform to the requirements of this specification after exposure to all combinations of the following conditions, for up to 16 months after delivery to the National Logistics Support Center.

TABLE III. Storage Environment

Condition	Range
Air Temperature	0 °C to 35 °C (32°F to 95°F)
Air Relative Humidity	15 % to 75 %
Dust	1 million particles per cubic meter, various sizes
Altitude	- 50.0 m to 3.0 km, MSL
Radiation	Sunlight through office windows and ultraviolet light from office fluorescent lights

3.2.2 Preparation and Release Environment Conditions.

Aerological sounding balloons shall meet the requirements of this specification after exposure without packaging to all combinations of the following conditions during preparation, inflation, and release during both day and night:

TABLE IV. Preparation and Release Environment Conditions

Condition	Range
Air Temperature	-60.0 °C to +50.0 °C
Air Relative Humidity	5.0 % to 100.0 % (condensing)
Dust	1 million particles per cubic meter, various sizes
Altitude	-50.0 m to 3.0 km, MSL
Surface Wind Speed	GP balloons: 0.0 to 25.0 Knots
Hydrometeors (liquid and frozen precipitation)	GP balloons: 0 to 150.0 mm per hour
Maximum Icing Conditions	GP balloons: 0.0254 cm in 6 minutes.

3.2.3 Flight Environment Conditions.

Aerological sounding balloons, when configured for a flight, shall meet the requirements of this specification when exposed to all combinations of the following conditions during both day and night flights:

TABLE V. Flight Environment Conditions	
Condition	Range
Air, Pressure	3.0 hectopascals (hPa) to 1070.0 hPa where the minimum air pressure for a particular balloon designation is limited to the actual atmospheric pressure at the minimum specified bursting altitude
Air, Temperature	GP balloons: -70.0 °C to +50.0 °C
Air, Relative Humidity	0.0 percent to saturation
Dust	1 million particles per cubic meter, various sizes
Wind speed in flight	0 meters/second (m/s) to 165 m/s
Hydrometeors (liquid and frozen precipitation)	GP balloons: 0.0 mm to 150.0 mm per hour
Maximum Icing Conditions	GP balloons: 0.0254 cm in 6 minutes.
Radiation and Atmospheric Chemistry	Exposure during flight to all radiation and chemicals occurring in the atmosphere up to the minimum specified burst altitude over the USA, US Possessions, and adjoining ocean areas.

3.3 Flight Performance Requirements.

3.3.1 Flight Preparation.

Aerological sounding balloons shall meet the requirements of this specification after being inflated and tied off with a cotton string. The aerological sounding balloon shall be adequately sealed with the amount of inflation gas specified by the Contractor to meet the requirements of this specification when the cotton string is tied with a tension 25 lbs to 35 lbs.

3.3.2 Inflation Gas.

Performance requirements of this specification shall be met when hydrogen gas only, natural gas only, or helium gas only is used to inflate the balloon. The maximum inflation gas required shall not exceed 2.7 cubic meters in the ambient conditions of 25°C and 1013.25 hPa. The preparation documentation shall specify the nozzle lift required (for helium, hydrogen and natural gas) to meet the specification requirements in the release conditions identified in TABLE V.

TABLE VI. Specific Release Conditions for Nozzle Lift		
Condition #	Parameter	Range
1	Moderate Breeze*	11 to 16 knots*
2	Fresh Breeze*	17 to 21 knots*
3	Strong Breeze* (limited)	22 to 25 knots*
4	Light Rain** (liquid and frozen)	Less than 2.8 mm per hour**
5	Moderate Rain** (liquid and frozen)	2.8 – 7.62 mm per hour**
6	Heavy Rain**(liquid and frozen)	More than 7.62 mm per hour**

* Beaufort Scale
** Federal Meteorological Handbook No. 1 (FCM-H1-2005)

3.3.3 Preparation

Preparation in the environmental conditions identified in 3.2.2 shall include all tasks necessary to cause the aerological sounding balloon to be ready-to-release. One person shall be able to successfully and routinely accomplish both preparation and launch of the balloon in the environment of 3.2.2. The preparation instructions shall include, but not be limited to removing the aerological sounding balloon from its individual packaging required by 5.1.1; placing the aerological sounding balloon on the Aerological Sounding Balloon Nozzle Assembly (FIGURE 1) and held by the Aerological Sounding Balloon Clamping Tool (FIGURE 2); identifying the recommended nozzle lift for the conditions identified in 3.3.2 and Table VI, for full specification compliance; tying or sealing the aerological sounding balloon to prevent inflation gas escape; and removing the aerological sounding balloon from the nozzle assembly.

3.3.4 Preparation-to-Release Time

Total preparation-to-release time in the environmental conditions identified in 3.2.2 for aerological sounding balloons supplied in accordance with this specification shall not exceed 45 minutes including preparation, inflation, and release. The time to inflate the aerological sounding balloon to the recommended amount of inflation gas shall not exceed 15 minutes. The aerological sounding balloons shall conform to the requirements of this specification when exposed to the environmental conditions identified in 3.2.2 during and after inflation and prior to release for up to 45 minutes.

3.3.5 Payload

Flight performance requirements shall be met with no more than a 600 gram payload. The flight payload for each flight may vary between 400 and 600 grams, based on the radiosonde type and other flight train materials.

3.3.6 Ascent Rates

The aerological sounding balloon ascent rates required by 3.3.7 of this specification shall be the average rate of ascent from the launch height to the actual burst altitude when the sample size is at least as large as required by 3.3.8. The maximum ascent rate from launch to 400.0 hPa shall not exceed 360.0 m/min (1181.1 ft/min).

3.3.7 Performance Percentage Requirements

At least 90.0 percent of the balloons shall meet the minimum burst altitude and minimum average ascension rate of TABLE VII when the sample size is at least as large as required by 3.3.8.

TABLE VII. Performance Percentage Requirements		
TYPE of Aerological sounding balloon	MINIMUM BURST ALTITUDE with respect to Mean Sea Level (km) that must be met by at least 90.0 % of balloons (km)	MINIMUM AVERAGE ASCENSION RATE Launch height to the actual burst altitude (meters/min)
GP20	20.0	275

3.3.8 Minimum Sample Size

The requirements of 3.3.6 and 3.3.7 shall be met when the sample sizes are at least as large as shown in TABLE VIII.

TABLE VIII. Minimum Sample Sizes	
Type of Test	Minimum Number of Valid Flight Tests
Initial Production Test	100
Lot Acceptance Test	40

If the number of failures occurring during a test makes it mathematically impossible to pass the test with the minimum sample size shown in TABLE VIII, then the Government reserves the right to discontinue testing and declare the balloons have failed the test. For example, during Production Acceptance tests, if 10 flights have been performed and 5 failures have occurred, the Government may discontinue testing since even if no further failures were to occur, the balloons would fail the test.

3.3.9 Balloon Launch

One person shall be able to launch the balloon, radiosonde, and parachute from outside a balloon launch shelter and from a balloon launch shelter in the environment of 3.2.2. The Contractor shall supply all special fixtures or machinery for launch, if required.

3.3.10 Federal Aviation Administration Requirements

The balloon shall conform to all requirements of Federal Aviation Requirement 101 (14 CFR 101.31 to 101.39).

3.3.11 Dispersal of Inflation Gas

The inflation gas of the extensible balloon shall be released to the atmosphere.

3.4 Material

3.4.1 Electrostatic Control

Aerological sounding balloon material shall not contribute to, or permit the buildup of, electrostatic charges on the aerological sounding balloon surface. An antistatic agent, if required, applied to the balloon meeting

3.0 requirements is not prohibited

3.4.2 Uniformity

The aerological sounding balloons shall be of uniform texture and have a consistent aerodynamic cross-section during flight.

3.4.3 Dusting

Aerological sounding balloons manufactured using construction techniques requiring powder dusting shall have uniform dusting with a non-toxic, dry, powdered material which effectively prevents adhesion of the material. The powder shall not react with hydrogen, helium, or Natural Gas. The powered material shall not contribute to the formation of an electrostatic charge during preparation, release, or flight.

3.4.4 Inflation Gas

Aerological sounding balloons shall not react with hydrogen, natural gas, or helium inflation gas. Aerological sounding balloon materials shall not expel or emit any chemical that will react with hydrogen, natural gas, or helium gas.

3.5 Physical Requirements

Aerological sounding balloons physical features shall meet the requirements set forth below.

3.5.1 Maximum Aerological Sounding Balloon Size

The size of the aerological sounding balloons supplied under this specification shall not exceed the following limits when fully inflated at 850 $\pm$10 hPa pressure.

TABLE IX. Maximum Aerological Sounding Balloon Size

BALLOON DESIGNATION	MAX INFLATED VERTICAL SIZE (meters)	MAX INFLATED HORIZONTAL SIZE (meters)
GP20	2.44	1.98

Vertical measurements do not include the neck-size requirements of 3.5.2.

3.5.2 Neck

Aerological sounding balloons shall be supplied with a neck securely attached to the body. The neck shall be at least 15.24 cm long and have an inside diameter between 3.073 cm to 4.52 cm. The neck shall not rip, shred, perforate, or separate from the balloon body when filled or removed from the nozzle. The edge (free end) of the neck of the aerological sounding balloon shall be free of nicks or folds. Aerological sound balloons shall be supplied with a body securely attached to the neck. When inflated the body shall conform to the dimensions in TABLE IX

As an alternative, the Contractor may submit aerological sounding balloons having different tubular-neck dimensions than those listed, but shall then be required to supply alternate Aerological Sounding Balloon Nozzle Assemblies and Aerological Sounding Balloon Clamping Tools. The alternate Aerological Balloon Nozzle Assemblies shall use the same eye ring assembly and a brass inflation gas fitting as shown in Figure 1. The alternate Aerological Balloon Nozzle Assemblies shall not contribute to the formation of an electrostatic charge during preparation. The alternative Aerological Sounding Balloon Nozzle Assembly and Aerological Sounding Balloon Clamping Tool shall be supplied at no cost to the Government. The quantity of Aerological Sounding Balloon Nozzle Assemblies and Aerological Sounding Balloon Clamping Tools provided by the Contractor for Initial Production Testing shall be seven (7). After successfully completing the Initial Production Testing, the Contractor shall supply the alternate Aerological Sounding Balloon Nozzle Assemblies and Aerological Sounding Balloon Clamping Tools to the Government at no charge for 102 sites plus 10 spares.

The neck shall consist of one or more structures to meet the following functional requirements:

a) The neck shall provide a means to inject the inflation gas during preparation and then to prevent the loss of the inflation gas.

b) The neck shall provide an ergonomic handle for the person to hold the balloon during its release.

c) The neck shall transmit the restraining forces from the operator encountered during release in the winds of 3.2.2 to the rest of the balloon so that the balloon is not overstressed.

d) The neck shall provide a means to attach the cotton string connected to the flight train to the balloon.

3.5.3 Material

3.6 Manufacturing Requirements

3.6.1 Physical Percentage

At least 99.0 percent of the aerological sounding balloons of each lot shall meet the size requirements of TABLE IX. The mean arithmetic deviation of the actual weights of individual balloons in a lot shall not differ by more than 10.0 percent. No more than 1 percent of a lot shall have physical defects (e.g. pinholes, thin spots, patches, etc).

3.6.2 Configuration Control

No change shall be made in design, material chemistry, or production process without prior written approval by the Contracting Officer.

3.6.3 Environmental Impact

Aerological sounding balloons required by this specification and factory equipment used to produce aerological sounding balloons required by this specification shall not cause exposure to hazardous substances exceeding the recommended limits provided in United States Center for Disease Control, Agency for Toxic Substance and Disease Registry, Minimal Risk Levels for Hazardous Substances in all of the following circumstances:

a) Aerological sounding balloon preparation.

b) Flight of the aerological sounding balloons.

c) Contamination of ground water and streams after flight due to leaching of chemicals from the aerological sounding balloon.

d) Ingestion of aerological sounding balloon remains in the ground water and in water bodies.

e) Disposal of aerological sounding balloons.

f) Exposure of Government employees visiting the factory(s) in preparation for, during, and after manufacture and test of aerological sounding balloons.

All other environment exposure limits imposed by other Government or Government agencies having jurisdiction are also required as though textually incorporated herein. This includes, but is not limited to, those requirements to protect the factory employees and the community(s) where the factory(s) is (are) located. The Contractor shall supply Material Safety Data Sheet (MSDS) to the Contracting Officer with the Offer for the balloons and, if used, also for dusting material.

4 AEROLOGICAL SOUNDING BALLOON TESTING

4.1 First Article Tests.

Deleted.

4.2 Production Testing and Lot Acceptance

4.2.1 Procedures

Steps in a successful aerological sounding balloon manufacturing and production lot acceptance are:

a) The Contractor shall supply specification compliant aerological sounding balloons.

b) Production Lot Samples shall be selected per 4.2.4.

c) Production Lot Samples shall be shipped at Contractor expense to NLSC at the address set forth in the contract.

d) Government test sites will perform tests as specified in APPENDIX A - J300-TP003 TEST PROCEDURES FOR AEROLOGICAL SOUNDING BALLOONS and supply the results to the Contracting Officer’s Representative (COR).

e) The COR will review test results and notify the Contractor of successful lot sample test completion.

f) The lot of aerological sounding balloons shall be shipped at Contractor expense to the NLSC in Kansas City, Missouri.

g) NSLC will inspect the shipping cartons for damage in accordance with 3.2.1.1 and the carton marking for compliance with 5.3.2 and Appendix B - Marking for Shipment.

h) NSLC will accept the lot of aerological sounding balloons F.O.B. NLSC. (NLSC will send a receiving report to the COR).

i) The Contractor shall send the invoice to the COR.

4.2.2 Inspection and Tests

The Government will perform inspection and tests on production lots of aerological sounding balloons on a random basis to determine compliance with this specification. In lieu of performing random inspection and testing of production lots, the Government may use aerological sounding balloon field performance data to verify compliance with specification requirements. The Government reserves the right to perform additional tests and the right to limit or eliminate tests without prior notification to the Contractor.

4.2.3 Production Lots

Aerological sounding balloons shall be accepted in lots. A production lot shall:

a) Not exceed 7,800 aerological sounding balloons;

b) Contain only aerological sounding balloons manufactured during a 14 consecutive day period; and

c) Be of the same type and designation. Individual aerological sounding balloon packaging and marking shall be in accordance with APPENDIX B - MARKING FOR SHIPMENT.

4.2.4 Selection of Production Lot Samples

Aerological sounding balloons of each production lot will be selected by a Government representative or, for the convenience of the Government, may be selected by the Contractor at the Contracting Officer's direction prior to placing individual aerological sounding balloon boxes into cartons. The aerological sounding balloons chosen for test shall be evenly distributed throughout the production lot.

Once selected, all samples will be forwarded to Government aerological sounding balloon test sites. The Production Samples from acceptable lots will be considered as partial delivery on the contract if the production lot is accepted.

The Contractor shall provide the COR documentation addressing use, handling, storage, preparation, and inflation requirements with both hydrogen only and helium only for each aerological sounding balloon contract.

5 PACKAGING REQUIREMENTS

5.1 Packaging

5.1.1 Individual Aerological Sounding Balloons

All aerological sounding balloons supplied in accordance with this specification shall be individually packaged in a manner that will ensure compliance with this specification. A simple method shall be provided for opening the individual packages without the use of a sharp instrument.

5.1.2 Cartons

5.1.2.1 Production Samples

Individually packed aerological sounding balloons shall be packaged in a manner ensuring compliance with this specification including 3.2.1.1 and 3.2.1.2.

5.1.2.2 Production Shipments to the NLSC

The individually packed aerological sounding balloons shall be shipped to the Government in cartons. An exterior carton of GP type balloons shall contain 30 individually packaged aerological sounding balloons of a specific lot. Intermediate cartons may be provided. The weight of a fully packed exterior carton shall not exceed 31.75 kg (70.0 lbs) and the girth plus length shall not exceed 2.74 m (108 in) to comply with postal regulations. All shipments exceeding ten cartons shall be transported on pallets and have plastic stretch wrapped over the cartons. All cartons shall be packaged in a manner that will ensure compliance with this specification when delivered to NLSC.

5.2 Item Lot Number and Month of Manufacture

The individually packaged aerological sounding balloons shall be shipped to the Government with a production lot number identified with the NSN and ASN. The production lot number shall be marked on the outside of the individually packaged aerological sounding balloon or on printed material contained within the packaging. The production lot numbers assigned to aerological sounding balloons shall conform to the code pattern WWWWWWWW-MMYYYY. Up to eight alphanumeric characters shall be traceable by the Contractor to their manufacturing operations, for example:

LOT: 260801DB-02-2010
or
LOT: DB26-801-02-2010
or
LOT: 801-02-2010

5.3 Marking of Shipment

Each exterior, intermediate, and unit carton shall be clearly marked to identify its contents.

5.3.1 Production Test Samples

The boxes shall be marked as required in 5.3.2 to include the aerological sounding balloon type and designation, the Contractor, and the production lot number. Production Test Samples in cartons shall be shipped to NLSC at the address set forth in the contract.

5.3.2 Production Shipments to NLSC

Once the COR has sent the Contractor a letter certifying successful lot sample test completion, the Contractor shall ship the production lot of individually packaged aerological sounding balloons in cartons to the NLSC marked as required below and in APPENDIX B - MARKING FOR SHIPMENT.

5.3.2.1 Exterior Container Markings

The exterior container shall be marked as shown in APPENDIX B except as described herein.

Upper Left Corner

- National Stock Number
- "CAGE" followed by the manufacturer's cage code.
- "Balloon" followed by the balloon designation
- "ASN:" followed by the agency stock number
- "30 EA" (GP20 balloons)
- "WT" followed by the exterior package weight in pounds. "CU" followed by the exterior package volume in cubic feet.
- Lot Number
- "Manufacture Date" followed by the date of manufacture in
  - month (spelled out) and year.
- "Warranty End Date" followed by the month (spelled out) and year 16 months following the expected delivery date to NLSC but not before step e of 4.2.1. (See Note).

NOTE: The warranty period starts with the actual delivery to NLSC. Since there is uncertainty as to when a shipment will arrive at NLSC, a one-month error will be tolerated in the marked date.

Lower Left Corner

- The contract number.
- The name of the Contractor.
- The Contractor's city, state, and zip code.

The bar codes shall show:

- The national stock number
- The contract number
- The Contractor's cage code

5.3.2.2 Intermediate and Unit Container Markings

The intermediate and unit containers, if used, shall be marked as shown in APPENDIX B MARKING FOR SHIPMENT.

6 WARRANTY

6.1 Warranty Period

The warranty period shall start on the actual delivery date of the balloon to NLSC and last for 16 months.

6.2 Visible Defects

All aerological sounding balloons exhibiting out-of-box defects (e.g. pinholes, thin spots, patches, etc) will be replaced at the Contractor's expense provided the Contractor is notified of the defect within the warranty period.

6.3 Defect Warranty

Should the flight test performance percentage of aerological sounding balloons for any production lot, at any individual or group of NWS site(s), fall below the requirements of Section 3.3.6 or TABLE VII within the 16- month warranty period, the COR will notify the Contractor of a defect. The Contractor may elect to have the lot (or a portion thereof) returned for inspection. The return shall be at the Government's discretion and at the Contractor's expense. The COR will supply the Contractor with information on the failure rate and other data that may be available. After evaluation, the Contractor may elect to either repair the remaining units or replace them. In either case, the balloons shall then be submitted for production lot acceptance in accordance with 4.2.

For warranty claims, the number of failed balloons must be at least as great as shown below:

TABLE X. Defect Warranty

PASS RATIO	MINIMUM NUMBER OF VALID TEST FLIGHTS
If the pass ratio less than 50.0 percent	20
If the pass ratio is greater than 50.0 percent but less than 80.0 percent	40
If the pass ratio is greater than 80.0 percent, but less than 90.0 percent	100

If the number of failures during a test makes it mathematically impossible to pass the test with the minimum sample size shown in above, then the Government reserves the right to suspend or terminate testing and declare the balloons are UNSATISFACTORY. For example, if 13 flights have been performed and 11 failures have occurred, the Government may discontinue testing since the balloons would fail the test should no further failures were to occur.

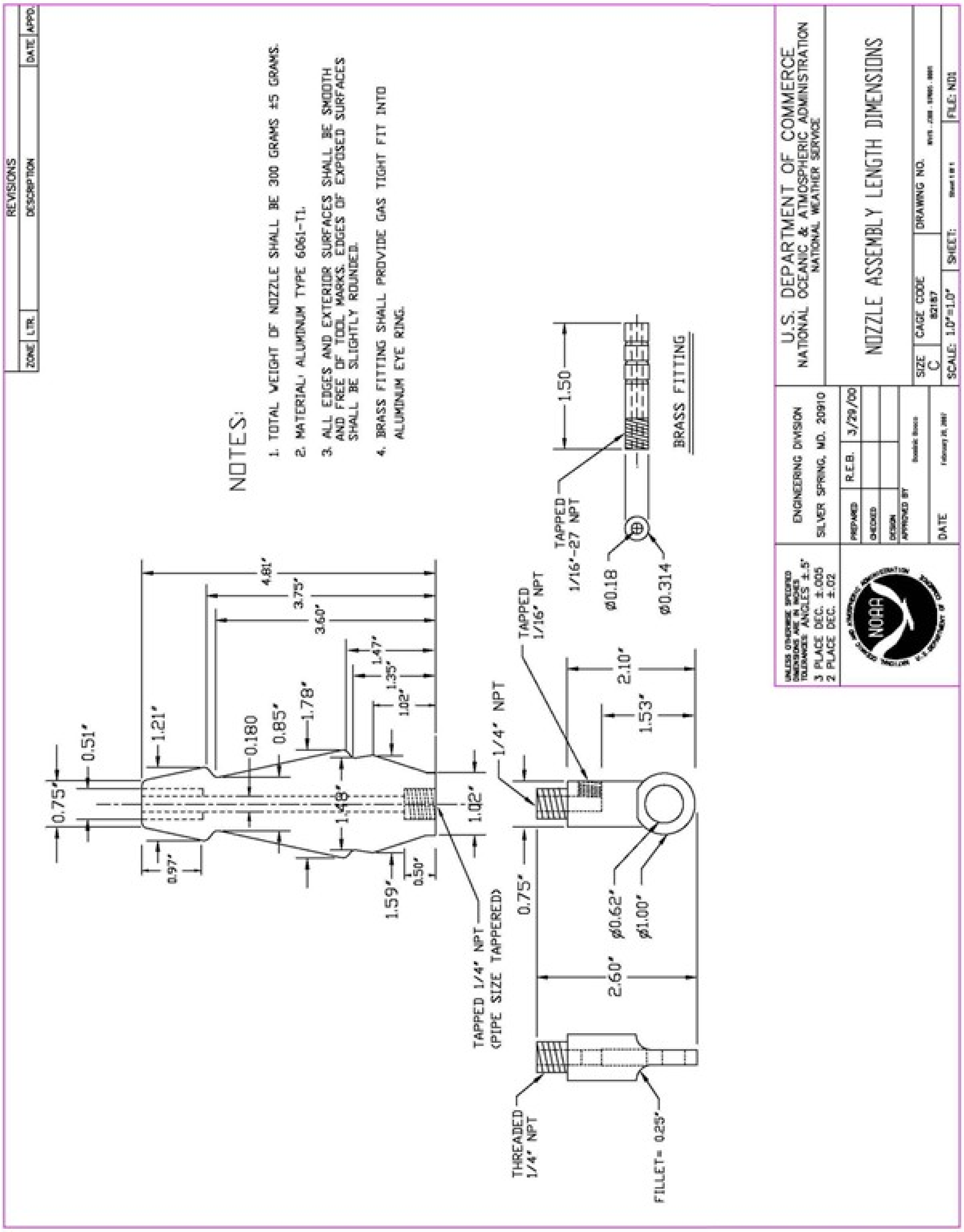

FIGURE 1. **Nozzle Assembly Length Dimensions**

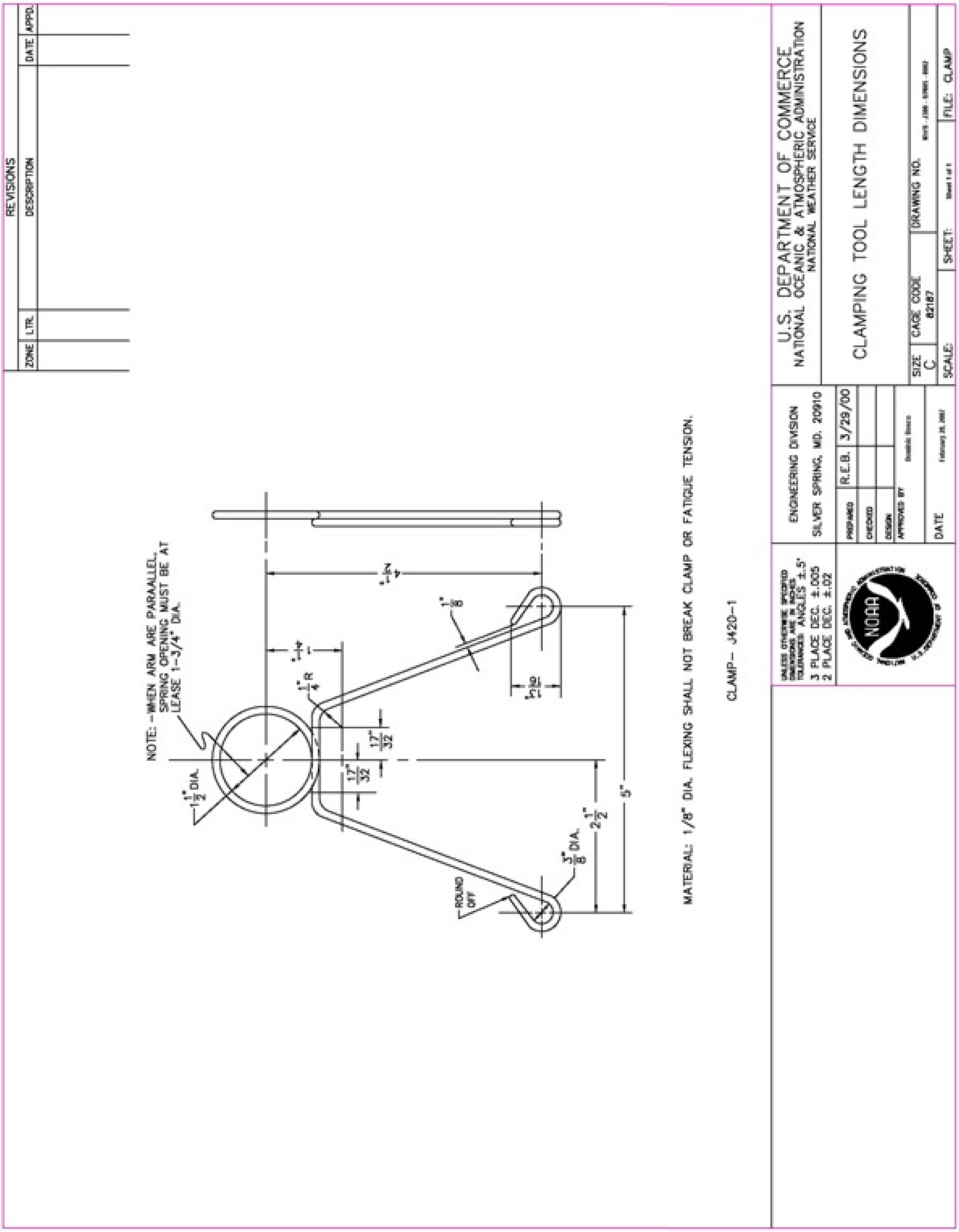

FIGURE 2. Clamping Tool Length Dimensions

Appendix to: J300-SP005
Revision L Ver. 0.1
January 15, 2016

NATIONAL WEATHER SERVICE

ENGINEERING and ACQUISITION BRANCH

Test Procedures for AEROLOGICAL SOUNDING BALLOONS

Appendix A
to
Specification No. J300-SP005

U.S. DEPARTMENT OF COMMERCE
NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION
NATIONAL WEATHER SERVICE
Silver Spring, Maryland 20910-3283

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TEST PROCEDURES FOR AEROLOGICAL SOUNDING BALLOONS

1 INTRODUCTION

This test plan outlines the requirements for free-flight, single-envelope, freely-expanding aerological sounding balloon testing. Aerological sounding balloons are used to carry radiosondes through the atmosphere to measure profiles of pressure, temperature, humidity, wind speed and wind direction. The National Weather Service (NWS) uses various types and designations of aerological sounding balloons to obtain these measurements. Radiosonde soundings are usually made at 00 UTC and 12 UTC, which, depending on longitude could be day-time or night-time conditions. The NWS flies over 60,000 aerological sounding balloons throughout its network of 92 stations and supplies balloons internationally. The NWS upper-air network extends south from Barrow, AK (71N) to Pago Pago, Samoa (14S) and west from Caribou, ME (68W) to Koror, Palau (134E).

1.1 Balloon Designations

Aerological sounding balloons are designated as General Purpose (GP) type and in accordance with their minimum bursting height in kilometers (km) above sea level.

1.2 Definitions

Valid Flight	For the purpose of balloon requirements and testing, a valid flight is one where either a) the balloon meets its performance requirements regardless of the cause of the flight termination or b) a properly launched flight is terminated by balloon burst and the flight environment conditions are within the requirements of this specification regardless of whether the balloon has met its performance requirements.
Invalid Flight	For the purposes of balloon requirements and testing, an invalid flight is one where either a) the radiosonde or ground station fails prior to the balloon reaching its minimum burst height or b) the balloon struck some object in the launch process or c) the flight environment conditions are outside the requirements of this specification.
Flight Test Performance Percentage	The flight test performance percentage is 100 times the ratio of the number of balloons meeting their performance requirements to the total number of valid flights.
Free Lift	Free lift is the number of grams of lift available over and above the number of grams of lift required by a balloon to support the weight of a complete radiosonde train.

Nozzle Lift	Nozzle lift is the number of grams of free lift plus the grams of lift required by a balloon to support the weight of a complete radiosonde train excluding the weight of the balloon. The radiosonde train not including the weight of the balloon consists of the radiosonde, parachute, light stick, dereeler or train regulator, and twine.
Gross Lift	Gross lift is the nozzle lift plus the grams of lift required to support the weight of the balloon.
Nominal	Test samples performance is in accordance with these requirements.
Suspended	Deficiencies have been encountered that prevent continuation of the test, but are not severe enough to cause termination. The Contracting Officers Representative (COR) or Test Review Committee (TRC) will determine if a test should be suspended. A potential contractor shall have no more than four (4) weeks to correct the deficiencies and resume the test. Otherwise, a termination test status may result.
Termination	Results from test samples that have failed one or more tests with such a severity of deficiencies that to continue testing would not be in the Government's interest. Only the COR will make this determination and report the findings to the CO.

2 METHODOLOGY FOR TEST AND EVALUATION

The methodology employed in this test plan will determine whether the Production Samples comply with NWS specifications by inspection, static tests, and flight tests of performance characteristics. Evaluation criteria will match the requirements given in their respective specification sections.

Specific specification paragraphs are shown for most of the tests of this procedure. The paragraphs shown are envisioned to be those where a deficiency revealed by the test would most likely be. A deficiency could also be found with other specification requirements.

2.1 Methodology for Production Test and Evaluation

2.1.1 Prerequisites for Production Tests

Aerological sounding balloons of each production lot will be selected by a Government representative or, for the convenience of the Government, may be selected by the Contractor at the Contracting Officer's direction prior to placing individual aerological sounding balloon boxes into cartons. The aerological sounding balloons chosen for test shall be evenly distributed throughout the production lot. Once selected, all samples will be forwarded to Government test sites. The Production Samples from acceptable lots will be considered as partial delivery on the contract if the production lot is accepted.

Documentation provided by the balloon vendor addressing the use, handling, storage, preparation, MSDS, and inflation requirements of hydrogen, natural gas and helium for each aerological sounding balloon type and designation.

Aerological Sounding Balloon Nozzle Assemblies and Aerological Sounding Balloon Clamping Tools shall be provided separately with the initial production lot samples if the alternative Aerological Sounding Balloon Nozzle Assembly and Aerological Sounding Balloon Clamping Tool are proposed.

2.1.2 Production Tests at Government Sites

The Government will conduct inspections and flight tests of Production Test Samples of all aerological sounding balloon lots.

2.1.2.1 Production Inspection at Government Sites

Upon receipt of the Production Test Samples, a physical inspection will be made of all aerological sounding balloon lots during flight preparation to determine compliance with specification sections:

3.2.1	Transportation and Storage Environment Conditions
3.2.2	Preparation and Release Environment Conditions
3.2.3	Flight Environment Conditions.
3.3.1	Flight Preparation
3.3.2	Inflation Gas
3.3.3	Preparation
3.3.4	Preparation-to-Release Time
3.3.5	Payload
3.4.1	Electrostatic Control
3.4.2	Uniformity
3.4.3	Dusting
3.4.4	Inflation Gas
3.5.1	Maximum Aerological Sounding Balloon Size
3.5.2	Neck
5.1.1	Individual Aerological Sounding Balloons
5.1.2.1	Production Samples
5.2	Item Lot Number and Month of Manufacture
5.3	Marking of Shipment

2.1.2.2 Production Evaluation

During flight preparation, the Government test participant shall note any physical deficiency with the aerological sounding balloons and include the notes with the flight data sheet.

2.1.2.3 Production Flight Tests at Government Sites

The Government will conduct a series of flight tests, day and night, of Production Test Samples of all aerological sounding balloon lots.

The aerological flight train and radiosonde will be prepared according to established NWS procedures. Aerological sounding balloon shall be prepared in accordance with the documentation submitted by the Contractor. The balloon will be inflated to the lift in grams as specified by the Contractor's instructions for 25 °C and 1013.25 hPa compensated for presented conditions. The total payload will be no more than 600 grams.

A surface observation will be made within ten minutes prior to release. Physical inspection and flight performance remarks will be recorded. A number of flights will be released in light to moderate precipitation, if possible. A number of flights will be released in surface winds greater than 15 kts, if possible.

After aerological sounding balloon release, the Upper Air ground system will track the radiosonde until burst. The Upper Air computer system will be used to compute the height of burst, as well as the ascent rate.

2.1.2.3.1 Production Flight Tests

The Government will prepare and conduct flight tests of Production Test Samples of all aerological sounding balloon types and designations to determine compliance with Specification Sections:

3.2.2	Preparation and Release Environment Conditions
3.2.3	Flight Environment Conditions.
3.3.5	Payload
3.3.6	Ascent Rates
3.3.7	Performance Percentage Requirements
3.4.1	Electrostatic Control
3.4.2	Uniformity
3.4.3	Dusting
3.4.4	Inflation Gas
3.5.1	Maximum Aerological Sounding Balloon Size
3.5.2	Neck
3.6.1	Physical Percentage

The Government reserves the right to request and test additional lot samples at operational sites located in the arctic or the tropics or to test aerological sounding balloons using precision radar tracking.

2.1.2.4 Production Evaluation

During flight preparation or the test flight, the Government test participant shall note any deficiency with the aerological sounding balloons.

2.1.3 Production Tests by the COR

During inspections and tests of Production Test Samples of all aerological sounding balloon lots, the COR shall note any deficiency with the aerological sounding balloons. In order for the aerological sounding balloons to be considered ACCEPTABLE during inspection, laboratory test, environmental tests, and flight tests, the aerological sounding balloons must meet the specification requirements. Additionally, in order for the aerological sounding balloons to be considered ACCEPTABLE during flight preparation and flight test, at least 90.0 percent of the aerological sounding balloon flights by the COR and at Government test sites must meet the specification requirements.

If aerological sounding balloons fail to meet the specification requirements, the aerological sounding balloon lot is considered UNSATISFACTORY. If at any time the performance of the aerological sounding balloons fails to meet the specification requirements, the COR may suspend further inspections and tests and forward all data to the Contracting Officer for resolution.

2.1.4 Aerological Sounding Balloon Production Test Schedule

The approximate duration for inspections and tests is ten workdays for each type of aerological sounding balloon lot. This estimate needs to be multiplied by the number of aerological sounding balloon lots for the total time required. The initial Production Test duration can be longer and is dependent on environmental conditions.

3 QUALITY ASSURANCE

3.1 Independence of Tests

During Production Lot Test and Evaluation, Production Lot Sample inspections and flight tests shall be conducted separately and independently on each aerological sounding balloon lot from each Contractor. Lot inspection and flight-test data shall not be compared with the results of other aerological sounding balloon lot data. Inspections and flight-test results shall not be released to other than the COR without the written permission of the Contracting Officer.

3.2 COR Reporting

During Production Lot Test and Evaluation, the COR will report the inspection and flight-test data results to the Contracting Officer and in accordance with the contract to the Contractor. Sample inspection and flight-test data shall be compared with the Specification requirements only. Production lot inspection and flight test-data shall not be compared with the results of other aerological sounding balloon data. Inspection and flight-test results leading to Production Lot rejection shall not be released to other than the Contracting Officer.

NATIONAL WEATHER SERVICE
LOGISTICS BRANCH

INSTRUCTIONS TO COMMERCIAL VENDORS FOR MARKING AND BAR CODING SUPPLIES FOR SHIPMENT TO THE NATIONAL LOGISTICS SUPPORT CENTER KANSAS CITY, MISSOURI

Appendix B
to
Specification No. J300-SP005

U.S. DEPARTMENT OF COMMERCE
NATIONAL OCEANIC AND ATMOSPHERIC ADMINISTRATION
NATIONAL WEATHER SERVICE
Silver Spring, Maryland 20910-3283

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Instructions to Commercial Vendors for Marking and Bar Coding Supplies for Shipment to the National Logistics Support Center, Kansas City, Missouri

Basic Requirements

Product and contract/purchase order information markings shall be placed on containers for items delivered to the National Logistics Support Center (NLSC). These markings shall contain information and be placed in accordance with requirements of MIL-STD-129M, Marking for Shipment and Storage, and MIL-STD-1189B, Standard Department of Defense Bar Code Symbology, with exceptions allowed as described below, or as contained in the contract or ordering document. The cited MIL-STD standards are available from the following source:

Defense Printing Service Detachment Office
Department of the Navy
700 Robbins Avenue
Philadelphia, PA 19111-5094

found at:

http://www.defenselink.mil/pubs/about.html

The information provided below is a synopsis of those MIL-STD requirements deemed important to logistics management requirements at the NLSC. Note: bar-code requirements are applicable unless they are specifically identified as excluded in the contract or purchase order.

A General Description of Requirements

The MIL-STDs contain requirements for content and placement of markings on containers of items delivered under contract or purchase order. These requirements include text markings of product, contract and vendor information, as well as select bar coded information.

Marking requirements differ for exterior containers and other levels of packing--unit packs, intermediate containers. An exterior container is defined as a container designed to protect unit packs and intermediate containers and their contents during shipment and storage. It can be a unit pack or an intermediate container, or contain a combination of unit packs and intermediate containers. An exterior container may or may not be used as a shipping container. An intermediate container is defined as a wrap, box, or bundle containing unit packs of identical items. A container which meets the definition as an intermediate container, but which is used as a shipping container, shall be marked as an exterior container. Any container which includes two or more National Stock Number's (NSN) shall be marked as an exterior container. Different purchase orders or contracts shall not be consolidated in a single container or pallet.

B. Text Information

Text information shall include product identification and Contractor information. These two types of information are described separately below, for each of the possible types of containers.

B.1. Identification markings on unit packs, intermediate containers, and unpacked items

In the order listed below, the following information is required unless specifically exempted in the procurement contract or order, or shown as optional.

NSN. NSN shall include spaces or dashes and any prefix or suffix shown in the contract or requisition.

CAGE Code and Part Number (PN). The letters "CAGE" and either "PN" or "P/N" shall be used to identify this information. The PN specified in the contract (if any) shall be used; if none is specified, the PN of the actual manufacturer or the PN assigned by the company awarded the contract. No PN is required if none is assigned to the item.

Item Description or Nomenclature. A short description of the product identified by the NSN. If specified in the contract or procurement document, the Agency Stock Number shall be shown on the next line(s), for example:

ASN: K220-3A3

Quantity (QTY) and Unit of Issue (UI). The quantity is the number of units of issue in the marked unit pack or intermediate container. The unit of issue is the standard or basic quantity indicated in the procurement contract or order as the minimum quantity issued, e.g., quart, liter, dozen, foot, meter, each, gross, etc. A nondefinitive UI shall be accompanied by a quantitative expression, such as:

"1 ROLL (100 FT)."

Item Lot Number and Month of Manufacture. The individually packaged aerological sounding balloons shall be shipped to the Government with a production lot number identified with the NSN and ASN. The production lot number shall be marked on the outside of the individually packaged aerological sounding balloon or on printed material contained within the packaging. The production lot numbers assigned to aerological sounding balloons shall conform to the code pattern WWWWWWWW-MM-YYYY. Up to eight alphanumeric characters shall be traceable by the Contractor to their manufacturing operations, for example:

LOT: 260801DB-02-2010
or
LOT: DB26-801-02-2010
or
LOT: 801-02-2010

Contract number or Purchase Order number, and line item number. When shelf-life markings are specified in the contract, purchase order, purchase description, specifications, material standards, and other procurement documents, they shall include the shelf-life expiration date, and be shown below the product description information on unit packs, intermediate containers, and unpacked items.

Hazardous Material Identification information shall be included in accordance with MIL-STD 129-M, and standards referenced therein. A Material Safety Data Sheet (MSDS) shall be included with each shipment.

An item assigned a serial number shall have that number applied to the unit pack and to the intermediate container preceded by the abbreviation "SER NO." The serial number data shall be shown directly below the product identification markings. When unit packs bearing consecutive serial numbers are packed in an intermediate container, only the first and last numbers in the series shall be shown. When not in sequence, each number shall be listed on the container. When there is inadequate space on the container for the marking of multiple serial numbers, a list of serial numbers shall be included. One copy of the serial number list shall be placed inside the container and one copy shall be placed in the packing list envelope. The identification-marked side of the container shall be marked:

"SERIAL NUMBER LIST INSIDE."

Figure B1 illustrates the content and placement of product identification markings on unit pack and intermediate containers.

B.2. Identification and contract markings on exterior containers

When necessary, due to a number of varying NSN's being shipped on the same contract at the same time, an exterior container may be required to consolidate several different unit packs and/or intermediate containers. In addition, an intermediate container may be utilized as a shipping container, and shall be marked as an exterior container. The following identification and contract information shall be marked on exterior containers in the order listed.

B.2.1 Exterior Containers Containing a Quantity of a Single NSN

If the exterior container contains one or more unit packs or one or more intermediate packs of the same NSN, the exterior container shall be marked with the following product description and contract information:

Identification Information

NSN , in the manner described in B.1 above.

CAGE Code and PN, in the manner described in B.1 above.

Item Description or Nomenclature. If specified in the contract or procurement document, the Agency Stock Number shall be shown on the next line, for example:

ASN: K220-3A3

Quantity and UI, as described in B.1. The quantity should reflect the total number of units of issue packed in the exterior container.

Gross Weight and Cube. The capital letters "WT" and CU" shall precede the gross weight and cube numerals. The gross weight shall be expressed in pounds, rounded up to the nearest pound. The cubic volume shall be calculated from the overall length, width, and height dimensions of the exterior shipping container. Irregular, cylindrical, or round items shall be considered to be rectangular solids. The volume shall be expressed in cubic feet, expressed in decimals rounded up to the nearest tenth of a cubic foot.

When shelf-life markings are specified in the contract, purchase order, purchase description, specifications, material standards, and other procurement documents, they shall include the shelf-life expiration date, and be shown below the product description information on exterior containers. NOTE: Any inspection/test date and other shelf-life markings, when required, shall be applied below the identification markings and as specified in MIL-STD-129M.

All required hazardous material markings shall be shown on the exterior container. For information on the additional marking and labeling requirements for hazardous materials, see MIL-STD-129M and other standards referenced therein.

Items assigned a serial number shall have that number applied, either in text format or bar coded, to the exterior container, preceded by the abbreviation "SER NO." The numbers shall be shown directly below the above product identification markings. When unit packs bearing consecutive serial numbers are packed in an exterior container, only the first and last numbers in the series shall be shown. When not in sequence, each number shall be listed on the container. When there is inadequate space on the container for the marking of multiple serial numbers, a list of serial numbers shall be included with the shipment.

One copy of the serial number list shall be placed inside the container and one copy shall be placed in the packing list envelope. The identification-marked side of the container shall be marked "SERIAL NUMBER LIST INSIDE."

Contract Information

Contract or Purchase Order Number, and Line Item Number. Where applicable, also include delivery order number, modification number, or lot number.

Name and Address of the Contractor (including nine-digit zip code). The street address is optional. When supplies are shipped from a sub-contractor; only the name and address of the company awarded the contract shall be used.

Figure B2 illustrates the content and placement of identification and contract markings on exterior containers which contain a quantity of a single NSN.

B.2.2 Exterior Containers of multiple NSN's

When containers of items include more than a single NSN, the container shall be marked with the following information, in the order listed, as shown on the next page:

Identification Information

Include on the first line the word "MULTIPACK." If the contained items are in support of a system identified in the order or contract, the system shall be identified on this line, e.g., "MULTIPACK FOR NEXRAD SYSTEM."

The Gross Weight and Cube.

In addition to any shelf-life markings, the words "CONTAINS SHELF-LIFE ITEMS" shall be placed on the next line(s).

Warranty Information. The words "WARRANTED ITEMS INSIDE" shall be placed on the next line if contained items are covered by a warranty.

Hazardous Material Identification markings, as specified in MIL-STD-129M and references contained therein, shall be placed on multi-pack containers.

Other special markings. Caution markings shall be applied as required (e.g., FRAGILE, arrows, hazardous warning labels, electrostatic discharge sensitive devices, etc.).

Items assigned a serial number shall have that number applied to the exterior container, preceded by the abbreviation "SER NO." The numbers shall be shown directly below the product identification markings listed in the previous paragraphs. When unit packs bearing consecutive serial numbers are packed in an exterior container, only the first and last numbers in the series shall be shown. When not in sequence, each number shall be listed on the container. When there is inadequate space on the container for the marking of multiple serial numbers, a list of serial numbers shall be included with the shipment.

One copy of the serial number list shall be placed inside the container and one copy shall be placed in the packing list envelope. The identification-marked side of the container shall be marked "SERIAL NUMBER LIST INSIDE."

Contract Information

Contract or Purchase Order Number, and Line Item Number. Where applicable, also include delivery order number, modification number, or lot number.

Name and Address of the Contractor (including nine-digit zip code). The street address is optional. When supplies are shipped from a sub-contractor; only the name and address of the company awarded the contract shall be used.

C. Shipping label markings

Shipping address markings or shipping labels shall be in accordance with accepted carrier practices and Department of Transportation regulations, or as otherwise prescribed in the contract or order. Address marking requirements of MIL-STD 129M are NOT included as part of requirements for shipment to NLSC, except that the Standard shall serve as guidance for placement of shipping labels on containers.

When multiple shipping containers are included in a shipment, the markings on each container shall include the serialized container number and the total number of containers in the shipment, e.g., 1 of 5.

All shipments to the NLSC shall be made to the following address:

DOC/USDA
National Logistics Support Center
1510 East Bannister Road, Bldg. 1
Kansas City, MO 64131-3009

D. Bar Code Markings

Unless otherwise specified in the contract or order, bar code markings are required on all containers and loose or unpacked items. Bar codes are to be of Code 3 of 9 symbology, generally of medium density (3.0 to 9.4 characters per inch). The extended 3 of 9 character set is NOT to be utilized. All bar codes and the human readable interpretation (HRI) shall meet the specifications contained in MIL-STD 1189B, Standard Department of Defense Bar Code Symbology. The HRI shall be an exact interpretation of the bar code data and shall not contain spaces or dashes or start/stop characters. The preferred location for the HRI is below the bar code markings, while the optional location is above the bar code markings.

D.1. Unit Packs and Intermediate Containers

Unit packs and intermediate containers shall contain a bar code of the NSN. This bar code shall consist of the basic 13 data characters. Prefixes and suffixes to the stock number, as well as spaces and dashes, shall not be bar coded. The HRI shall include only the 13 basic data characters. Dashes, spaces or start/stop characters shall not be shown in the HRI.

When a requirement for bar-coded serial numbers is specified in the procurement contract or order, an item assigned a serial number shall have the bar-coded serial number applied to the unit pack and to the intermediate container. The bar codes shall be located directly below the product identification markings preceded by the abbreviation "SER NO." The letters "SER NO" shall not be bar coded. When more than five serial-numbered items are in an intermediate container, two serial number lists shall be provided. One list, which is to be placed inside the container, shall contain a bar code for each serialized item. The second list shall be placed in the packing list envelope; bar coding of the second serial number list is optional. The words "SERIAL NUMBER LIST INSIDE" shall be marked on the product identification-marked side of the container.

Figure B3 illustrates the marking placement and bar-code requirements for unit packs and intermediate containers.

D.2. External Containers

Information to be bar coded on exterior containers which contain a single NSN shall include:

(1) NSN. The NSN shall be bar-coded as specified in D.1.

(2) Contract/Delivery Order or Purchase Order Number. The bar-coded contract/delivery order or order number shall consist of up to 15 characters, as prescribed in the contract or order.

(3) CAGE Code of the company awarded the contract. The CAGE code shall consist of five characters.

(4) Contract Line Item Number (CLIN) when used. The CLIN shall consist of six characters, including zero fillers placed to the left of the number (e.g., 0001AB).

NOTE: For exterior containers containing more than a single NSN, such as multipacks, bar-code markings on the exterior container shall be limited to the contract or order number. However, unit packs and intermediate containers which are included in the exterior container shall be bar-coded as described in D.1.

HRI information shall be preferably shown under the bar code, and shall include only the bar-coded data characters; start/stop characters shall not be shown.

In addition, when a requirement for bar-coded serial numbers is specified in the procurement contract or order, all items assigned a serial number shall have the bar-coded serial number applied to the exterior container. The bar-codes shall be located directly below the product identification markings and be preceded by the abbreviation "SER NO;" the letters "SER NO" shall not be bar coded. When more than five serial-numbered items are contained in an exterior container, two serial number lists shall be provided. The first list, which is to be placed inside the container, shall contain a bar code for each serialized item. The second list shall be placed in the packing list envelope; bar coding of the second serial number list is optional. The words "SERIAL NUMBER LIST INSIDE" shall marked on the product identification-marked side of the container.

Figure B4 illustrates the placement of markings and bar-code requirements for external containers.

Figure B5 illustrates the layout format acceptable for bar-code markings on exterior containers.

E. Markings and Marking Materials

Markings shall be by stenciling, stamping, machine printing, or labeling (using preprinted labels). In addition, hand printing may be used on packs and containers for marking any of the following: serial numbers, quantity per pack, date of pack, total pieces, and weight and cube. Hand printing must be clear and legible. Abbreviations associated with these markings, such as "SER NO," "WT," and “CU" may be hand printed.

The lettering for all markings on all sizes of containers shall be in capital letters of equal height, shall be clearly legible, and shall be proportionate to the available marking space on the container. When marking space permits, stenciled or pre-printed markings shall be not less than one-fourth of an inch nor more than 1-inch in height. Unless otherwise specified in the procurement or contract document, identification, contract data, and address markings shall be not less than .095 inch (approximately three thirty-seconds of an inch).

If labels for exterior containers are not inherently waterproof, they shall be waterproofed by coating the entire outer surface of the label with a transparent, waterproofing material, such as spar varnish, acrylic coating compound, or other protective coating. Transparent tape may also be applied over markings to provide waterproof protection; however, readability of bar codes must be assured for this type of waterproof protection.

Paper labels shall be made of sized white paper stock having a smooth finish and a minimum basis weight of 20 pounds. Labels shall be of a water-resistant grade of paper, film, fabric, or plastic and shall be coated on the unprinted side with a water-insoluble, homogeneous, pressure-sensitive, permanent type adhesive. The adhesive shall adhere to metal, plastic, or fiberboard surfaces under high and low temperatures associated with expected shipping and warehousing conditions. Labels shall have a finish capable of withstanding normal handling during shipment and storage and shall be suitable for printing and writing on with ink without feathering or spreading. The applied label must remain securely in position under anticipated conditions of handling, shipment and storage.

F. Packing Slip Documentation

Each stand-alone container (defined as the outermost container of a shipment) shall be individually marked with a packing slip. Additionally, all stand-alone packages shall contain an internal packing slip to ensure proper documentation in the event of loss of the external packing slip. Each packing slip shall be marked within the following information:

1. National Stock Number
2. CAGE code and PN
3. Item Description or Nomenclature
4. Agency Stock Number, if identified in the contract or order
5. Quantity in the container and Unit of Issue
6. Contract or Purchase Order Number and Line Item Number. (Where applicable, include delivery order number.)
7. Shelf-Life Expiration Date
8. Hazardous Material Identification
9. Serial Number(s), if warranty tracking or specialized control of stock is required.
10. Container number and total number of containers included in the shipment (e.g., 1 of 5)

If desired, the contract or purchase order number and CAGE code may be marked in the heading of the packing slip in order to avoid repetition. The packing slip shall be securely attached to the end or side of the container, and sealed in a waterproof container conforming to MIL PPP-E540, Class 1. They shall be secured to the exterior of the palletized load or containers in the most protected location with tape, or pressure sensitive adhesive backing applied to containers other than wood. On wood containers, tacks or staples shall be used.

G. Pallet Requirements

Shipments shall be palletized to the maximum extent practical. When a shipment is palletized, the pallet size must be 44 inches by 40 inches (1.12 m by 1.016 m). The maximum stacking height shall not exceed 72 (1.83 m) for the GP type balloons. The shipment shall be palletized in numerical container order, shrink-wrapped and double strapped with metal or nylon straps across both directions.

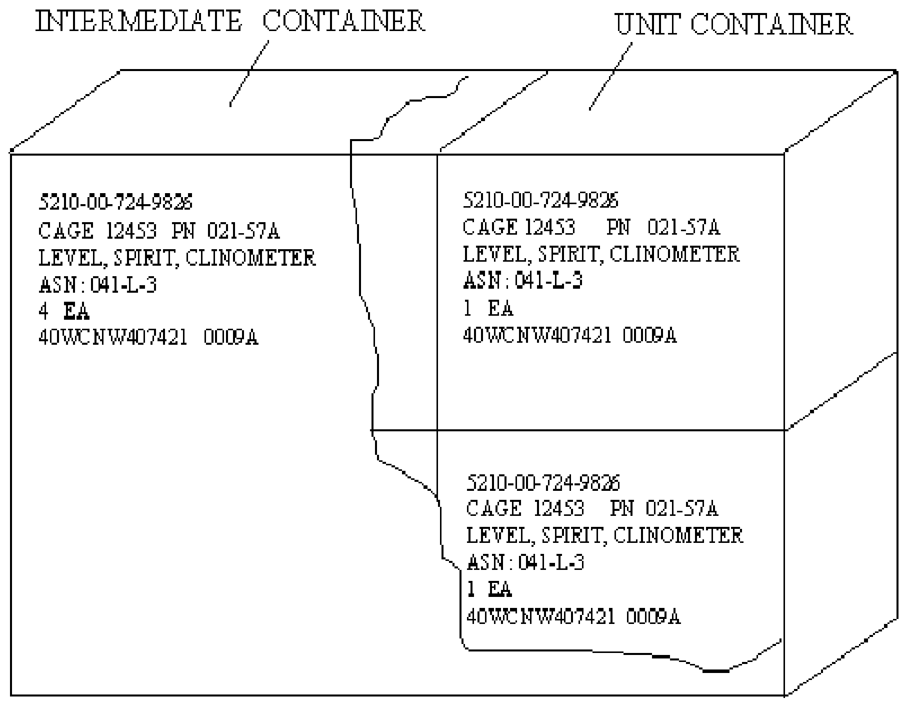

Figure B1 – Identification and Contract Markings for Unit Pack and Intermediate Containers

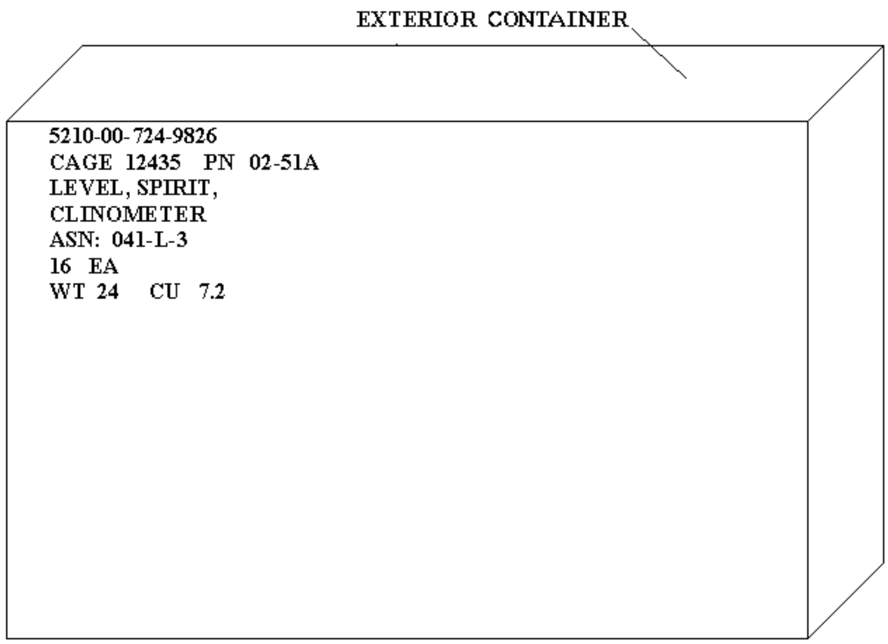

Figure B2 – Typical Exterior Container Text Markings

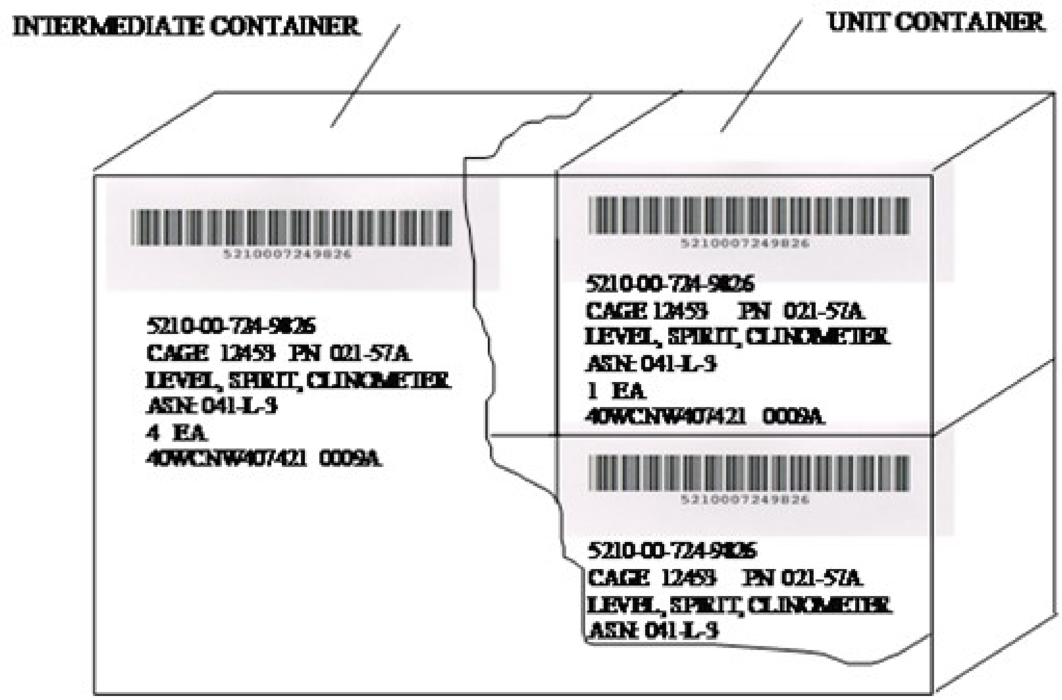

Figure B3 – Barcode Marking on Unit Packs and Intermediate Containers

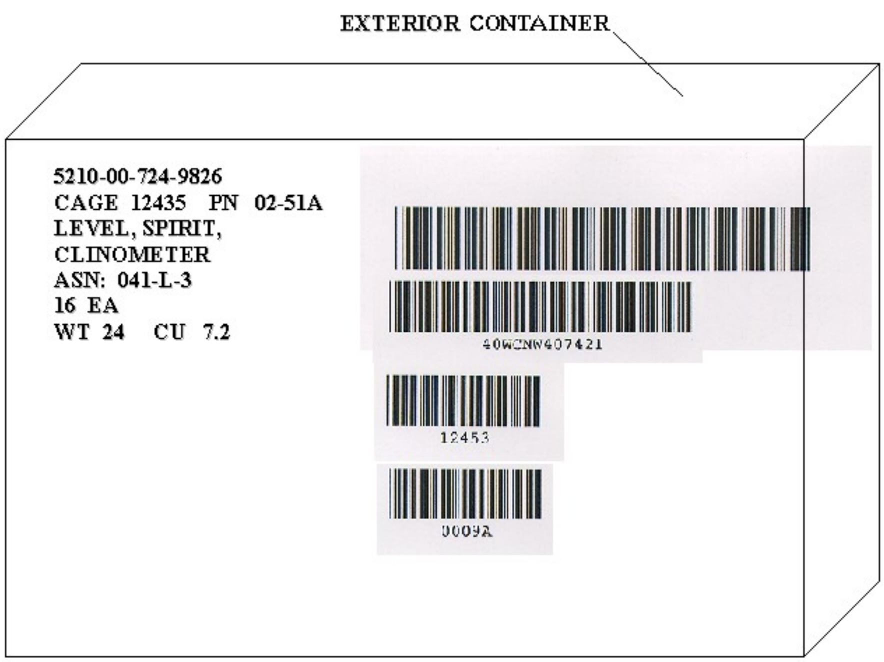

Figure B4 – Typical Exterior Container Text and Bar Code

Figure B5 - Acceptable bar-code layout formats for exterior containers.